# Bayesian Statistical Modeling: Gibbs Sampling, Data Augmentation, and Variable Selection

**MSC IN APPLIED STATISTICS**
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# Gibbs Sampling for Pareto Type I Distribution

We are given population data for $n = 51$ US states in 2012. We assume that the populations $X_1, \dots, X_n$ form a random sample from a Pareto Type I distribution with cumulative distribution function

$$P(X \leq x|a,b) = 1 - \left(\frac{b}{x}\right)^a, \quad x > b$$

and probability density function

$$f(x|a,b) = \frac{ab^a}{x^{a+1}}, \qquad x > b, \;\; a,b > 0$$

The prior distribution for $(a,b)$ is taken as $\pi(a,b) \propto 1$ for $a,b > 0$. The goal is to estimate the parameters $a,b$ using Gibbs sampling.

## 1. Conditional posterior distributions

For the data $x_1, \dots, x_n$ the likelihood is

$$f(x|a,b) = a^n b^{an} \left(\prod_{i=1}^{n} x_i^{-(a+1)}\right) \times I(minx_i > b)$$

Distribution of $a$ given $b$:

$$\pi(\alpha|b,x) \propto a^n b^{an} \left(\prod_{i=1}^{n} x_i^{-(a+1)}\right)$$

Taking logarithms and simplifying:

$$\pi(\alpha|b,x) \propto a^n exp\left[an\, lnb - (a+1)\sum_{i=1}^{n} lnx_i\right]$$

$$\pi(\alpha|b,x) \propto a^n exp\left[-\alpha \sum_{i=1}^{n} \ln\left(\frac{x_i}{b}\right)\right] \sim Gamma\left[n+1, \sum_{i=1}^{n} \ln\left(\frac{x_i}{b}\right)\right]$$

which is the density of a Gamma distribution with shape $n+1$ and rate $\sum_{i=1}^{n} \ln(x_i/b)$. Distribution of $b$ given $a$:

$$\pi(b|a,x) \propto b^{an} \times I(0 < b < x_{(1)})$$

Integrating yields

$$\pi(b|a,x) \propto \frac{an+1}{x_{(1)}^{an+1}} b^{an}, \qquad 0 < b < x_{(1)}$$

Setting $b = x_{(1)}V$ we obtain $V \sim \text{Beta}(an+1,1)$. Hence

$$f_V(y) = \frac{an+1}{x_{(1)}^{an+1}} \left(x_{(1)}y\right)^{an} x_{(1)}$$

$$f_V(y) = (an+1)y^{an}$$

$$V \sim Beta(an+1,1)$$

With these conditional distributions we can implement a Gibbs sampler to simulate from the joint posterior of $(a, b)$.

## 2. Results

In our implementation we used $M = 10000$ total iterations and $B = 1000$ as burn-in. Based on the simulated chain we computed the posterior estimates (sample means) of $a$ and $b$, as well as the 95% credible intervals (the 2.5% and 97.5% quantiles of the simulated values). The results are summarized in the following table:

| Parameter | Estimate | 2.5% | 97.5% |
|---|---|---|---|
| a | 5.337831e-01 | 3.941618e-01 | 6.898090e-01 |
| b | 5.433618e+05 | 4.921658e+05 | 5.622891e+05 |

*Table 1: Posterior summaries for parameters a and b*

The estimates are obtained as the means of the Gibbs samples, while the credible intervals reflect the posterior uncertainty about the parameters of the Pareto model.

# Data Augmentation for a Univariate Distribution

The goal is to apply Data Augmentation to simulate from the distribution

$$f(x) \propto \frac{x^3 e^{-5x}}{(x+3)^2}, \qquad x > 0$$

The presence of the term $(x+3)^{-2}$ in the denominator suggests introducing an auxiliary variable $w$ via the integral representation

$$\frac{1}{(x+3)^2} = \int_0^\infty w e^{-w(x+3)} dw$$

which follows from the exponential distribution. Hence the joint distribution of $(x, w)$ is

$$f(x,w) \propto x^3 e^{-5x} w e^{-w(x+3)}, \qquad x > 0, w > 0$$

$$f(x,w) \propto w x^3 e^{-(5+w)x-3w}$$

From this joint density we derive the full conditional distributions:

$$f(w|x) \propto w e^{-w(x+3)} \sim Exponential(x+3)$$

$$f(x|w) \propto x^3 e^{-(5+w)x} \sim Gamma(4, 5+w)$$

With these conditionals, the Gibbs sampler proceeds as follows:

In our implementation we used $M = 10000$ total iterations and $B = 1000$ as burn-in. From the generated chain, we estimated the expected value of $\frac{x+3}{x+2}$:

$$\hat{E}\left[\frac{x+3}{x+2}\right] = 1.3677$$

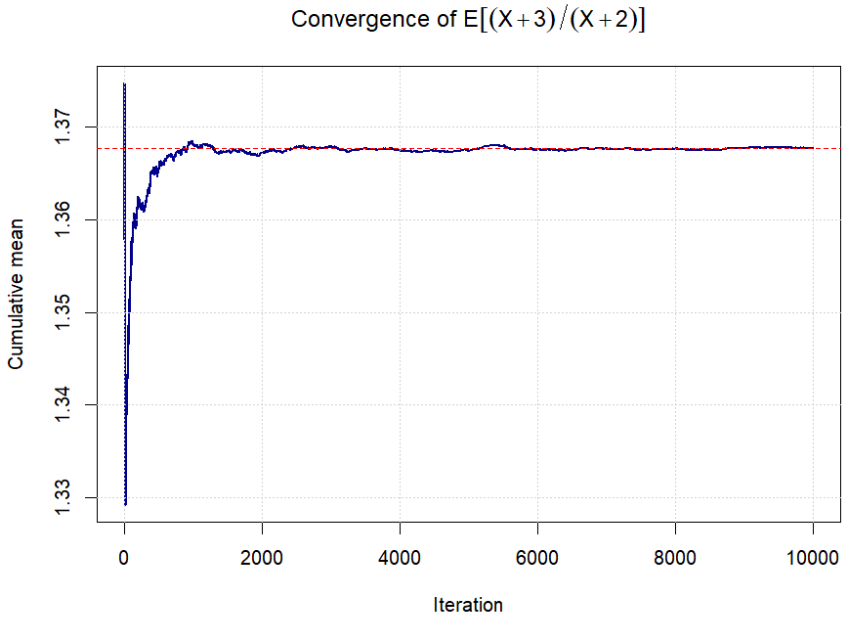


*Figure 1: Running mean of (x+3)/(x+2) versus iteration number.*

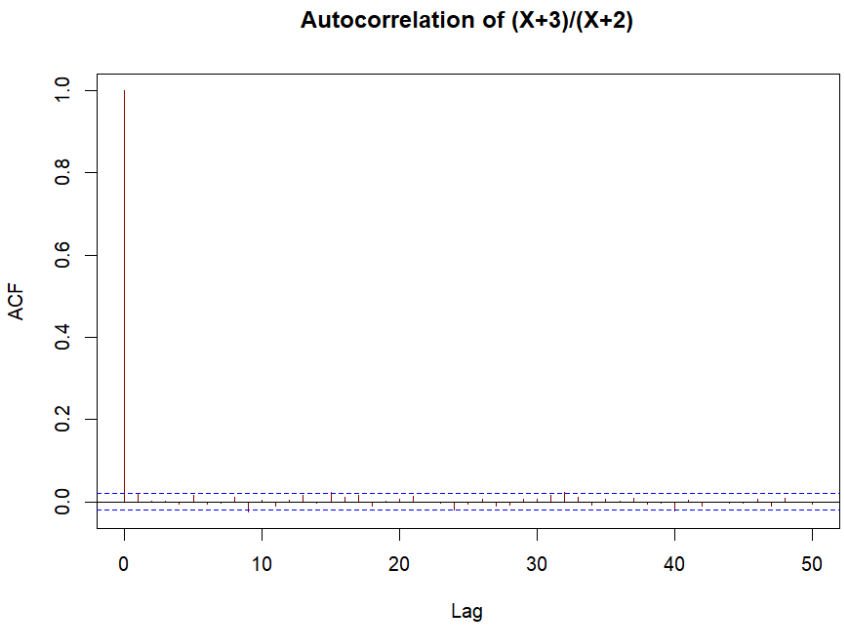


*Figure 2: Autocorrelation of the (x+3)/(x+2) chain as a function of lag.*

To assess convergence and mixing, we constructed:

- Trace plot of the running mean: showing the cumulative average of $\frac{x+3}{x+2}$ as a function of the iteration number. After the burn-in period, the value stabilizes around 1.3677.

- Autocorrelation function (ACF) plot: displaying very low autocorrelations for lags greater than 1, indicating good mixing and rapid convergence of the chain.

These results confirm that the Data Augmentation approach provides an efficient and straightforward implementation of the Gibbs sampler for the target distribution.

# Bayesian Analysis of Real Estate Prices with Variable Selection

The aim is to analyze data from 100 properties with the following variables:

- **price** (y): sale price (in thousand euros)
- **sqm** (x1): area (m²)
- **age** (x2): building age (years)
- **dist_center** (x3): distance from the center of Athens (km)

Three stages are applied: classical multiple linear regression, Bayesian multiple linear regression with non-informative priors, and a hierarchical Bayesian variable selection model. Predictions for new properties follow.

## 1. Classical multiple linear regression

The results (least squares estimates and p-values) are shown in the table:

| Variable | Coefficient | P-value |
|---|---|---|
| Intercept | -17.843 | 0.008 |
| Sqm | 1.472 | 0.000 |
| Age | -0.742 | 0.000 |
| Dist_center | -0.189 | 0.615 |

*Table 2: Coefficient estimates and p-values from classical multiple linear regression.*

At a significance level of 0.05, the variables sqm and age are statistically significant, while distance from the center is not.

## 2. Bayesian multiple linear regression (all variables)

The following priors are used:

- For the coefficients: $\beta \sim N_p(0, \tau_0^{-2}(X'X)^{-1})$with $\tau_0^2 = 10^{-5}$ (non-informative).
- For the precision $\tau^2 = 1/\sigma^2$: $\tau^2 \sim$ Gamma(0.005,0.005).

A Gibbs sampler is run with $M = 10000$ iterations and burn-in $B = 1000$. The posterior means and 95% credible intervals are:

| Variable | Estimate | 2.5% | 97.5% |
|---|---|---|---|
| Intercept | -17.855 | -31.055 | -4.973 |
| Sqm | 1.470 | 1.355 | 1.585 |
| Age | -0.741 | -1.017 | -0.471 |

| | | | |
|---|---|---|---|
| Dist_center | -0.185 | -0.931 | 0.555 |

*Table 3: Posterior means and 95% credible intervals for the coefficients in the Bayesian regression (all variables).*

The Bayesian estimates are almost identical to the classical ones, as expected due to the non-informative priors.

## 3. Hierarchical model with variable selection

Latent binary variables $w_j$ are introduced for each coefficient ($j = 1,2,3$). The priors:

- $w_j \sim Bernoulli(0.5)$
- Hyperparameters: $\gamma_0 = \delta_0 = \varepsilon = \zeta = 0.01$

A Gibbs sampler is used ($M = 10000$, burn-in 1000). The posterior inclusion probabilities for each variable are:

| Intercept | Sqm | Age | Dist_center |
|---|---|---|---|
| 1 | 1 | 1 | 0.007 |

*Table 4: Posterior inclusion probabilities for each variable.*

The most probable model is **model 4** (Intercept, Sqm, Age) with posterior probability **0.993**. For this model, the posterior means of the coefficients are:

| Intercept | Sqm | Age |
|---|---|---|
| -19.230 | 1.474 | -0.741 |

*Table 5: Posterior mean coefficients for the most probable model (Intercept, Sqm, Age).*

According to the most probable Bayesian model, the price of a property is positively affected by area and negatively by age. Holding other variables constant, each additional square meter increases the price by approximately 1.474 thousand euros, while each additional year of age decreases it by approximately 0.741 thousand euros (741 euros). In the following graph, we also show the convergence of the standardized means of the variables.

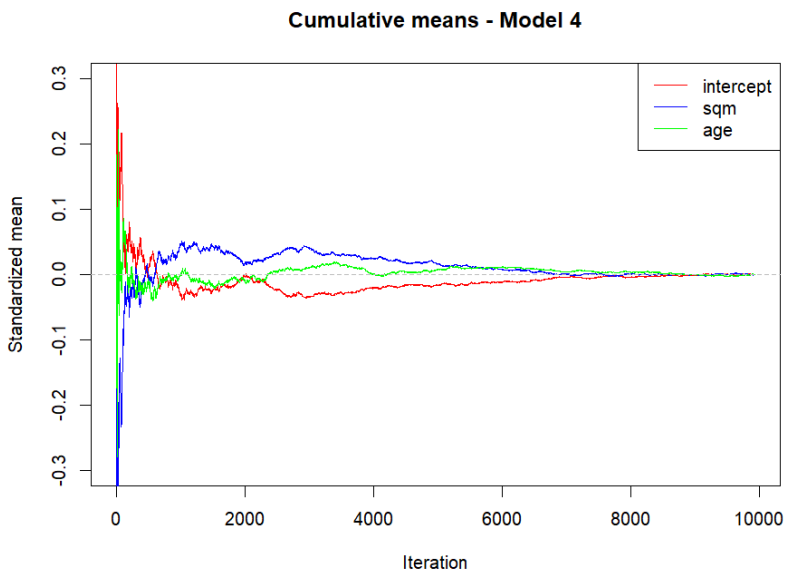


*Figure 3: Trace plot of the standardized means of the coefficients.*

## 4. Predictions for new properties

Three types of properties are given:

| Type | sqm | age | dist_center |
|---|---|---|---|
| Small downtown apartment | 55 | 25 | 1.5 |
| New suburban apartment | 85 | 5 | 8.0 |
| Large semi-urban apartment | 120 | 15 | 12 |

*Table 6: Characteristics of new properties for prediction.*

Prediction using the most probable model (Intercept, Sqm, Age):

| Type | Prediction | LCL | UCL |
|---|---|---|---|
| Small downtown apartment | 43.21 | 21.46 | 64.15 |
| New suburban apartment | 102.29 | 81.31 | 123.69 |
| Large semi-urban apartment | 146.50 | 124.83 | 168.06 |

*Table 7: Point predictions and 95% prediction intervals for the new properties using the most probable model.*

Prediction using model averaging (weighting all models by their posterior probabilities):

| Type | Prediction | LCL | UCL |
|---|---|---|---|
| Small downtown apartment | 43.25 | 21.34 | 64.87 |
| New suburban apartment | 102.29 | 81.50 | 123.56 |
| Large semi-urban apartment | 146.51 | 124.80 | 167.83 |

*Table 8: Point predictions and 95% prediction intervals using model averaging.*

The predictions and intervals are nearly identical, which is due to the dominance of model 4 (probability 99.34%). Model averaging effectively places almost all weight on this model. Although the results coincide, model averaging is theoretically more appropriate for the business because it incorporates model uncertainty, protects against the risk of choosing a wrong model, and provides more realistic prediction intervals for business decision-making.

The factors that significantly affect property prices are area (positive) and age (negative), while distance from the center has no statistically significant effect. The Bayesian estimates agree with the classical ones, but the Bayesian approach provides direct probabilistic interpretation and quantification of uncertainty through credible intervals and inclusion probabilities. The hierarchical model confirms that only area and age are necessary. Predictions from the most probable model and from model averaging give practically identical results, with model averaging being preferred for its incorporation of model uncertainty.